Discrete Structures

Set Laws: Proofs, Questions & Solutions

Set Laws: Reference Sheet (1/2)

- **Identity:** $A \cup \emptyset = A, \quad A \cap U = A$
- **Domination:** $A \cup U = U, \quad A \cap \emptyset = \emptyset$
- **Idempotent:** $A \cup A = A, \quad A \cap A = A$
- **Complement:** $A \cup A' = U, \quad A \cap A' = \emptyset$
- **Double complement:** $(A')' = A$
- **Commutative:** $A \cup B = B \cup A, \quad A \cap B = B \cap A$
- **Associative:** $(A \cup B) \cup C = A \cup (B \cup C)$ (same for $\cap$)

Set Laws: Reference Sheet (2/2)

- **Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$**
- **Distributive: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$**
- **De Morgan: $(A \cup B)' = A' \cap B'$, $(A \cap B)' = A' \cup B'$**
- **Absorption: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$**
- **Difference: $A - B = A \cap B'$**

This last law is the key: it turns a difference into ∩ and complement, so the other laws can be used.

How to Prove a Set Identity

- **Method 1: Use the laws (algebraic proof)**

  Start with one side. Apply ONE law per line and write its name. End at the other side.

  Start with the more complicated side. Rewrite $A - B$ as $A \cap B'$ first.

- **Method 2: Element (membership) proof**

  Show LHS $\subseteq$ RHS and RHS $\subseteq$ LHS using "$x \in \ldots$".

- **Method 3: Membership table**

  Check all in/out cases for the elements.

- **To DISPROVE: give one counterexample with real sets.**

Questions: Set Laws

Q1. Prove using laws: $(A \cap B) \cup (A \cap B') = A$

Q2. Prove using laws: $A - (B \cup C) = (A - B) \cap (A - C)$

Q3. Simplify: $(A \cup B) \cap (A \cup B')$

Q4. Prove using laws: $(A \cap B)' \cup B = U$

Q5. Prove using laws: $A \cup (B - A) = A \cup B$

Q6. Prove by elements: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Questions: Set Laws (continued)

Q7. Use a membership table to prove $(A \cup B)' = A' \cap B'$.

Q8. Simplify: $A \cap (A' \cup B)$

Q9. Is $A - (B - C) = (A - B) - C$ always true? Prove it or give a counterexample.

Solution Q1

(A ∩ B) ∪ (A ∩ B′)

= A ∩ (B ∪ B′)

Distributive law, taking A common.

= A ∩ U

Complement law: B ∪ B′ = U.

= A

Identity law: A ∩ U = A. [QED]

Check: A = {1,2,3}, B = {2}: {2} ∪ {1,3} = {1,2,3} = A ✓

Solution Q2

A − (B ∪ C) = A ∩ (B ∪ C)′

Difference law: X − Y = X ∩ Y′.

= A ∩ (B′ ∩ C′)

De Morgan.

= (A ∩ A) ∩ (B′ ∩ C′)

Idempotent: A = A ∩ A (we need A twice).

= (A ∩ B′) ∩ (A ∩ C′)

Commutative and associative: regroup.

= (A − B) ∩ (A − C)

Difference law again. [QED]

Solution Q3 and Q4

Q3: (A ∪ B) ∩ (A ∪ B′)

= A ∪ (B ∩ B′) » Distributive (used backwards)

= A ∪ ∅ » Complement law

= A » Identity law

Q4: (A ∩ B)′ ∪ B

= (A′ ∪ B′) ∪ B » De Morgan

= A′ ∪ (B′ ∪ B) » Associative

= A′ ∪ U » Complement law

= U » Domination law [QED]

Solution Q5

A ∪ (B − A) = A ∪ (B ∩ A′)

Difference law: B − A = B ∩ A′.

= (A ∪ B) ∩ (A ∪ A′)

Distributive law: A ∪ (B ∩ A′).

= (A ∪ B) ∩ U

Complement law: A ∪ A′ = U.

= A ∪ B

Identity law. [QED]

Check: A = {1,2}, B = {2,3}: {1,2} ∪ {3} = {1,2,3} = A ∪ B ✓

Solution Q6 (Part 1): LHS ⊆ RHS

Let $x \in A \cap (B \cup C)$.

So $x \in A$ and $x \in B \cup C$.

Definition of $\cap$.

So $x \in A$ and ($x \in B$ or $x \in C$).

Definition of $\cup$.

Case 1: $x \in B$. Then $x \in A \cap B$.

Case 2: $x \in C$. Then $x \in A \cap C$.

In both cases $x \in (A \cap B) \cup (A \cap C)$.

Being in either part puts x in the union.

Solution Q6 (Part 2): RHS ⊆ LHS

Let $x \in (A \cap B) \cup (A \cap C)$.

So $x \in A \cap B$ or $x \in A \cap C$.

Case 1: $x \in A \cap B$. Then $x \in A$ and $x \in B$, so $x \in B \cup C$.

Case 2: $x \in A \cap C$. Then $x \in A$ and $x \in C$, so $x \in B \cup C$.

In both cases $x \in A$ and $x \in B \cup C$.

So $x \in A \cap (B \cup C)$.

Both parts proved, so the two sets are equal [QED]

Equal sets = each is a subset of the other.

Solution Q7: Membership Table

A	B	A∪B	(A∪B)′	A′	B′	A′∩B′
1	1	1	0	0	0	0
1	0	1	0	0	1	0
0	1	1	0	1	0	0
0	0	0	1	1	1	1

1 = in the set, 0 = not in the set. The 4 rows cover every case. Column (A∪B)′ equals column A′∩B′ in every row, so the sets are equal [QED]

Solution Q8 and Q9

Q8: $A \cap (A' \cup B)$

$= (A \cap A') \cup (A \cap B)$ » Distributive

$= \emptyset \cup (A \cap B)$ » Complement law

$= A \cap B$ » Identity law

Q9: NOT always true. Counterexample:

$A = \{1,2,3\}$, $B = \{2,3\}$, $C = \{3\}$

$A - (B - C) = A - \{2\} = \{1,3\}$

$(A - B) - C = \{1\} - \{3\} = \{1\}$

$\{1,3\} \neq \{1\}$, so the identity fails.